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Editor-in-Chief
Journal of Complex Networks

Dear Editor-in-Chief,

Please consider the manuscript “Construction Sensitivity of a Shortest-Path Anisotropy Diagnostic in Geometric Graphs from Black-Hole Embeddings” for publication in the *Journal of Complex Networks*.

The paper studies how graph construction and matched-control choice affect a shortest-path multiplicity statistic in spatial geometric graphs. Schwarzschild/Flamm, Bardeen, and Hayward embedding surfaces are used as controlled benchmark geometries rather than as a basis for a new astrophysical prediction. The principal design is a paired ten-seed comparison across five graph/control protocols, supplemented by canonical Bardeen and Hayward parameter scans. The main contribution is methodological: it distinguishes reproducibility within a fixed protocol from sensitivity across constructions and does not claim a universal curvature measure.

An earlier version appeared as arXiv:2606.27248 and reported fixed-protocol and single-realization results. The submitted manuscript replaces that construction analysis with paired ten-seed audits, adds canonical Bardeen and Hayward calculations, and removes quantitative claims that did not survive multi-seed validation. It is a revised and extended version of the same project, not an unrelated manuscript.

OpenAI ChatGPT and Codex were used to assist with language editing, code organization, reproducibility checks, and preparation of the manuscript materials. The author reviewed and verified the text, code, analyses, figures, and numerical results and takes full responsibility for the contents of the work.

The manuscript is not under consideration elsewhere. I approve its submission to the *Journal of Complex Networks*.

Sincerely,

Ariadna Uxue Palomino Ylla
Department of Physics, Nagoya University